

PRATIK SAPKOTA

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|  [Pratiksapkota007](#) |

Lalitpur, Nepal

EDUCATION

• Kathmandu University

Bachelors of Electrical and Electronics Engineering (communication)

◦ GPA: 2.96

May 2023

Dhulikhel, Nepal

EXPERIENCE

• Wireless Engineer

Huawei Technologies Co. Ltd.

Oct 2023 - May 2025

Kathmandu, Nepal

- Managed configuration, integration and optimization of Huawei BSC6900, BSC6910, RNC6900 and RNC6910, improving **network stability** and **service availability**
- Commissioned BTS3900, NodeB and eNodeB sites including LTE rollout on legacy 2G and 3G layers, accelerating **coverage expansion**
- Executed modernization and capacity upgrades on more than 500 sites through 4T4R, 8T8R, SFN, CoMPT, RRU swaps and UBBP expansions, increasing **traffic handling efficiency**
- Optimized radio parameters for more than 100 sites including power, tilt, RET, AISU and AAU wide beam configurations, **enhancing user throughput and experience**
- Performed **software and version upgrades** for all the live NCELL sites, completing **KPI validation** and alarm clearance to prevent **performance degradation**
- Integrated **30+ new RAN sites** independently, **reducing rollout timelines** and accelerating project delivery
- Resolved multi domain faults by coordinating with transmission, core and OSS teams, **minimizing downtime** during incidents including major nationwide fiber cuts
- Supported on air readiness, field quality audits and **24 by 7 service assurance** during high profile national events in Nepal, maintaining **network performance commitments**
- Delivered remote and on site wireless support to **Huawei Sri Lanka operations**, ensuring successful **deployment and adherence** to implementation standards

• System Administrator Intern

Thakral One Nepal Pvt Ltd

June 2023 - Sept 2023

Kathmandu, Nepal

- Supported **virtualization deployment** for banking clients using VMware vSphere
- Assisted in **Windows Server installation**, configuration and **maintenance** for enterprise environments
- Participated in **server migration** and **Disaster Recovery setup** including RAID configuration and physical network connectivity
- Prepared monthly server **health reports** and provided on site **technical support** during deployment activities

• Academic Intern

Dish Media Pvt Ltd

Dec 2022 - Jan 2023

Lalitpur, Nepal

- Monitored live satellite broadcast signals to ensure service continuity for DishHome subscribers
- Supported preventive maintenance of DTH uplink, IRD systems, and Fiber TV headend equipment
- Conducted channel locking adjustments and antenna fine tuning to optimize signal quality and reduce error rates
- Assisted in diagnosing RF and digital transmission issues using signal analyzers and monitoring tools while applying knowledge of transport streams, conditional access systems, and transponder configurations

PROJECTS

- **Learning-Based Link Selection in Integrated Ground–Air–Space Networks (Simulation Study)** 2026
ns-3, C++, NetAnim [G]
 - Built an ns-3 simulation for learning-based link selection across terrestrial, UAV, and LEO links using network performance indicators to support adaptive handover decisions
 - Evaluated the impact of the selection policy on throughput, reliability, and handover frequency using C++ and NetAnim
- **Topography-Aware Predictive Beamforming and Handover for AI-RAN** 2026
ns-3, C++, NetAnim, Python [G]
 - Built an ns-3 digital twin for UAV-mounted cells and a ground UE with DEM-based terrain LoS/NLoS and a 100 ms step control loop.
 - Implemented a Python Gym + PPO RL pipeline with mock and bridge modes, and exported CSV/plot + TensorBoard artifacts for evaluation.
 - Developed an O-RAN-style xApp (FastAPI) exposing health + policy decision endpoints aligned with the RL action interface (handover target + beam phase).
- **Federated Learning for Network Anomaly Detection** 2025
Python, TensorFlow, NumPy, Pandas, Scikit learn [G]
 - Implemented a working federated averaging protocol across distributed clients, achieving up to 99.8% test accuracy under IID partition settings
 - Developed centralized and federated deep learning baselines for intrusion and anomaly detection on the UNSW NB15 dataset, including binary CNN and DNN architectures
- **ECG Myocardial Infarction Detection with ResNet** 2025
Python, PyTorch, NumPy, Pandas [G]
 - Processed ECG signals for noise removal, feature extraction, and detection of MI specific abnormalities using morphological analysis
 - Implemented ResNet18 based ECG classifier for four cardiac conditions (Normal, MI, PMI, HB), integrating signal processing to convert ECG signals into diagnostic images
- **LoRa-based Flood Monitoring System** 2023
Python, PennyLane, TensorFlow [G]
 - Established long-range communication; performed path-loss analysis
 - Designed a flood monitoring system with a cloud-based data acquisition system utilizing real-time communication

RESEARCH PUBLICATION

C=CONFERENCE, S=IN SUBMISSION, T=THESIS, J=JOURNAL

- [C.1] Atit Pokharel, Shashank Dahal, **Pratik Sapkota**, Bhupendra Bimal Chhetri (2023). **Electrocardiogram (ECG) Based Cardiac Arrhythmia Detection and Classification using Machine Learning Algorithms..** In Proceedings of the International Conference on Technologies for Computer, Electrical, Electronics and Communication (ICT CEEL 2023). October 13, Bhaktapur, Nepal.
- [J.1] Atit Pokharel, Sanskar Aryal, **Pratik Sapkota**, Prajun Trital, Rejina Karki,(2026). **PF-Caps: A Capsule Network-Based Federated Learning Framework for Personalized and Secure Learning in Healthcare** "In review" *IEEE Transactions on Medical Imaging*
- [T.1] Atit Pokharel , **Pratik Sapkota** , Dilip Sapkota , Shashank Dahal , Sulav Karki. **Performance Evaluation of LoRa Technology for Rural Connectivity: An Experimental Analysis in Nepal** . *BSc Thesis*, Department of Electrical and Electronics Engineering, Kathmandu University, Dhulikhel, Kavre, Nepal . Supervisor, Dr.Ram kaji Budhathoki. ([link](#))

SKILLS

- **Programming Languages:** Python, C++, Javascript, MATLAB, Linux
- **Machine Learning:** Tensorflow, Pytorch
- **Electronics:** Circuit Design, Proteus, Arduino
- **Wireless :** NS-3, NetAnim, iMaster MAE(Huawei Tool), Link Budgeting, System Design
- **Soft Skills:** Team Collaboration, Technical Communication, Research Assistance, Microsoft Office
- **Languages:** English (IELTS :8.0), Hindi, Nepali

CERTIFICATION

- Nepal Engineering License, Nepal Engineering Council